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CEN-431 - Computer Networks Lab

Design and Implementation of a Multi-Service Network Using Cisco Packet Tracer

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IP Addressing and Subnet Summary

Table 1: IP subnet and addressing summary.

Network / VLAN	Subnet	Hosts	Gateway / Purpose
Internet	9.9.9.0/24	254	Edge 9.9.9.1; NAT outside and Internet cloud
Branch-Core WAN	172.16.0.0/30	2	Branch 172.16.0.1; Core 172.16.0.2; WAN link
Edge-Core WAN	172.16.0.4/30	2	Core 172.16.0.5; Edge 172.16.0.6; WAN link
VLAN 3 (Access)	192.168.3.0/24	254	Gateway 192.168.3.1; access network
VLAN 10 (HR-HQ)	192.168.10.0/24	254	Gateway 192.168.10.1; main office HR
VLAN 20 (Finance-HQ)	192.168.20.0/24	254	Gateway 192.168.20.1; main office finance
VLAN 30 (IT)	192.168.30.0/24	254	Gateway 192.168.30.1; main office IT
VLAN 40 (HR-Branch)	192.168.40.0/24	254	Gateway 192.168.40.1; branch HR and wireless clients
VLAN 50 (Financial)	192.168.50.0/24	254	Gateway 192.168.50.1; branch financial users
VLAN 99 (Mgmt.)	192.168.99.0/24	254	Gateway 192.168.99.1; management network
Datacenter	192.168.1.0/27	30	Gateway 192.168.1.1; web and secondary DNS server
Server farm	192.168.60.0/24	254	Gateway 192.168.60.1; DHCP and primary DNS server
Edge loopback	10.1.1.1/32	1	Edge Loopback1; router identification
Branch loopback	3.0.0.1/32	1	Branch Loopback3; router identification

The usable-host values show the normal number of assignable addresses after excluding the network and broadcast addresses. The loopback addresses are /32 host routes, so each one represents a single usable host address.

3 Switching and VLAN Configuration

Switching was implemented using Cisco 2960-24TT switches. Instead of listing every repeated command, this report shows the important command pattern and explains how it was applied. The same access-port pattern was repeated for each user VLAN.

3.1 Main Office Switch Configuration

The **Main Office Switch** uses `spanning-tree mode pvst`. Interface Fa0/1 is the trunk toward the **Main Office Edge Router**, and it carries VLANs 3, 10, 20, 30, and 99. Access ports were assigned as follows: Fa0/2 to VLAN 10, Fa0/3 to VLAN 20, Fa1/4 to VLAN 30, Fa2/5 to VLAN 3, and Fa3/6 to VLAN 99.

```
spanning-tree mode pvst

interface FastEthernet0/1
  description Trunk_to_Edge_Router
  switchport mode trunk
  switchport trunk allowed vlan 3,10,20,30,99

interface FastEthernet0/2
  description HR-HQ
```

```
switchport mode access
switchport access vlan 10
spanning-tree portfast
```

Listing 1: Simplified main office switch command pattern.

The final access-port commands were repeated for VLAN 20, VLAN 30, VLAN 3, and VLAN 99 with the correct interface number, description, and VLAN ID.

3.2 Branch Switch Configuration

The **Branch Switch** also uses `spanning-tree mode pvst`. Interface Fa0/1 is the trunk toward the **Branch Router**. Interface Fa0/2 is assigned to VLAN 40 with description HR-BRANCH, Fa0/3 is assigned to VLAN 50 with description `financial`, and the wireless access point is connected as an access port in VLAN 40.

```
spanning-tree mode pvst

interface FastEthernet0/1
description Trunk_to_Branch_Router
switchport mode trunk
switchport trunk allowed vlan 40,50

interface FastEthernet0/2
description HR-BRANCH
switchport mode access
switchport access vlan 40
spanning-tree portfast
```

Listing 2: Simplified branch switch command pattern.

The same access-port command pattern was used for Fa0/3 in VLAN 50 and for the access point port in VLAN 40.

4 Routing Protocols & Inter-VLAN Routing

Router-on-a-stick routing was used on the main office edge router and on the branch router. Each VLAN is represented by a router subinterface configured with `encapsulation dot1Q`. DHCP relay was enabled with `ip helper-address 192.168.60.2` so that clients in different VLANs can reach the centralized DHCP server.

Dynamic routing was divided into two routing domains. The main office and core networks use EIGRP process 100. The branch side uses RIP. The **Core Router** performs mutual redistribution between EIGRP and RIP using explicit seed metrics to ensure redistributed routes are accepted.

4.1 Main Office Edge Router Configuration

The **Main Office Edge Router** connects to the core through Fa0/0 172.16.0.6/30, routes the main office VLANs through Fa0/1 subinterfaces, and connects to the Internet cloud through Eth0/0/0 9.9.9.1/24. The following example shows the pattern used for all main office VLAN subinterfaces.

```

interface FastEthernet0/1.10
  description VLAN_10_HR-HQ
  encapsulation dot1Q 10
  ip address 192.168.10.1 255.255.255.0
  ip helper-address 192.168.60.2
  ip nat inside

interface Ethernet0/0/0
  description NAT_Outside_to_Internet
  ip address 9.9.9.1 255.255.255.0
  ip nat outside

access-list 1 permit 192.168.10.0 0.0.0.255
access-list 1 permit 192.168.20.0 0.0.0.255
access-list 1 permit 192.168.30.0 0.0.0.255
access-list 1 permit 192.168.40.0 0.0.0.255
access-list 1 permit 192.168.50.0 0.0.0.255
ip nat inside source list 1 interface Ethernet0/0/0 overload

```

Listing 3: Simplified edge router inter-VLAN and NAT command pattern.

The same subinterface pattern was repeated for VLAN 3, VLAN 20, VLAN 30, and VLAN 99. The corrected NAT ACL also included the WAN subnets 172.16.0.0/30 and 172.16.0.4/30, plus the datacenter subnet 192.168.1.0/27.

4.2 Core Router Configuration

The **Core Router** connects the main office edge router, the branch router, the datacenter subnet, and the server farm subnet. Its most important routing role is redistribution between EIGRP and RIP.

```

router eigrp 100
  no auto-summary
  network 172.16.0.4 0.0.0.3
  network 192.168.1.0 0.0.0.31
  network 192.168.60.0 0.0.0.255
  redistribute rip metric 10000 0 255 1 1500

router rip
  version 2
  no auto-summary
  network 172.16.0.0
  redistribute eigrp 100 metric 3

```

Listing 4: Simplified core router route redistribution commands.

The EIGRP redistribution metric 10000 0 255 1 1500 provides a seed metric for RIP routes entering EIGRP. The RIP redistribution metric 3 provides a hop-count seed metric for EIGRP routes entering RIP.

4.3 Branch Router Configuration

The **Branch Router** uses RIP and provides inter-VLAN routing for VLAN 40 and VLAN 50. VLAN 40 is used by HR-Branch users and wireless clients, while VLAN 50 is used by financial users.

```
interface FastEthernet0/1.40
description VLAN_40_HR-BRANCH
encapsulation dot1Q 40
ip address 192.168.40.1 255.255.255.0
ip helper-address 192.168.60.2

interface FastEthernet0/1.50
description VLAN_50_financial
encapsulation dot1Q 50
ip address 192.168.50.1 255.255.255.0
ip helper-address 192.168.60.2

router rip
version 2
no auto-summary
network 172.16.0.0
network 192.168.40.0
network 192.168.50.0
```

Listing 5: Simplified branch router inter-VLAN and RIP commands.

5 Network Services

The project uses a centralized server model. The DHCP server and primary DNS server are located at 192.168.60.2 in the server farm subnet. A second DNS service is also configured on 192.168.1.2 in the datacenter subnet. DHCP relay is configured on router subinterfaces using `ip helper-address 192.168.60.2`, allowing clients from all VLANs to receive addresses from the correct pools.

5.1 DHCP Server Configuration

Table 2: DHCP pools configured on the server at 192.168.60.2.

Pool Name	Network	Gateway	Hosts	DNS Servers
vlan 3 access	192.168.3.0/24	192.168.3.1	254	192.168.60.2, 192.168.1.2
datacenter	192.168.1.0/27	192.168.1.1	30	192.168.60.2, 192.168.1.2
VLAN10	192.168.10.0/24	192.168.10.1	254	192.168.60.2, 192.168.1.2
VLAN20	192.168.20.0/24	192.168.20.1	254	192.168.60.2, 192.168.1.2
VLAN30	192.168.30.0/24	192.168.30.1	254	192.168.60.2, 192.168.1.2
VLAN40	192.168.40.0/24	192.168.40.1	254	192.168.60.2, 192.168.1.2
VLAN50	192.168.50.0/24	192.168.50.1	254	192.168.60.2, 192.168.1.2
serverPool	192.168.60.0/24	0.0.0.0	254	192.168.60.2, 192.168.1.2

The usable-host column shows the number of normal assignable host addresses in each pool. In practice, some of these addresses may be reserved for gateways, servers, or network devices.

5.2 DNS and Web Service

Two DNS services were configured for name resolution. The primary DNS server is 192.168.60.2, and the secondary DNS server is 192.168.1.2. The required DNS A record is:

`www.internal.com → 192.168.1.2`

The internal web server is placed in the datacenter subnet at 192.168.1.2. Clients from the main office, branch office, and wireless VLAN can resolve `www.internal.com` and reach the web service after routing and DHCP relay are configured correctly.

5.3 Wireless Service

The wireless access point is connected to the **Branch Switch** and bridged to VLAN 40. The access point was configured with SSID `Branch_Visitors`, WPA2-PSK using AES encryption, and channel 6. Wireless clients receive DHCP addresses from the VLAN 40 pool and use 192.168.40.1 as their default gateway.

6 Testing and Verification

Testing was performed after completing VLAN, trunking, routing, DHCP, DNS, NAT/-PAT, and wireless configurations. The following checks confirmed that the network operates as designed.

1. **Inter-VLAN pings:** A PC in VLAN 10 successfully pinged a PC in VLAN 30. This confirmed that the main office router-on-a-stick configuration on **R-EDGE** was routing between VLAN subinterfaces correctly.
2. **DHCP verification:** The command `ipconfig /renew` was executed on client PCs. Each client received the correct IP address, subnet mask, default gateway, and DNS server settings from its assigned pool. All VLAN clients received addresses from the correct DHCP pools hosted on 192.168.60.2, with DNS pointing to 192.168.60.2 and 192.168.1.2.
3. **End-to-end routing and NAT/PAT:** A branch PC successfully pinged the main office web server at 192.168.1.2 and the Internet-facing address 9.9.9.1. On the edge router, `show ip nat translations` confirmed that PAT was translating inside addresses through Ethernet0/0/0.
4. **Routing table verification:** The command `show ip route` on the **Core Router** showed EIGRP and redistributed RIP routes. The command `show ip route` on the **Branch Router** showed RIP routes learned from the core, including routes redistributed from EIGRP.
5. **Wireless verification:** Both laptops associated with SSID `Branch_Visitors`, received VLAN 40 DHCP addresses, and successfully reached internal resources such as `www.internal.com` as well as external resources through the edge router.

7 Challenges Encountered & Resolutions

Table 3: Implementation challenges and applied resolutions.

Challenge	Resolution
ACL typos such as 192.169.x.x prevented NAT for some internal subnets.	The NAT ACL was corrected to use the proper 192.168.x.x networks.
DHCP broadcasts were blocked at router boundaries.	The command <code>ip helper-address 192.168.60.2</code> was added on all router subinterfaces that serve client VLANs.
Route redistribution was missing seed metrics.	Seed metrics were added on the Core Router : <code>redistribute rip metric 10000 0 255 1 1500</code> for EIGRP and <code>redistribute eigrp 100 metric 3</code> for RIP.
VLAN description mismatch existed on branch router subinterfaces.	Descriptions were corrected to match the switch access VLANs: HR-BRANCH for VLAN 40 and financial for VLAN 50.

8 Pros and Cons / Lessons Learned

Pros

- **Scalability:** The VLAN-based structure allows new departments and access networks to be added with limited disruption.
- **Fast EIGRP convergence:** EIGRP improves convergence in the core and main office routing domain.
- **Centralized DHCP and two DNS services:** Central DHCP simplifies address management, while primary and secondary DNS improve name-resolution availability.
- **PAT security:** PAT hides internal client addresses when they access the Internet-facing network.
- **Wireless protection:** WPA2-PSK with AES provides stronger wireless access protection than open authentication.

Cons

- **Single point of failure:** The core router is a critical dependency for communication between the main office, branch, datacenter, and server farm.
- **Router-on-a-stick bottleneck:** Inter-VLAN traffic depends on router trunk links, which may become a bottleneck as traffic increases.
- **Manual ACL risk:** Manually written NAT ACLs are error-prone, as shown by the original 192.169.x.x typo.
- **No redundancy:** The topology does not include backup routers, redundant WAN links, or failover paths.
- **RIP overhead:** RIP periodic updates are acceptable for a small branch but are less efficient than modern routing protocols.

Lessons Learned

- NAT ACLs must be verified carefully because a single subnet typo can prevent Internet access.
- DHCP relay using `ip helper-address` is essential when DHCP clients and servers are separated by routers.
- Route redistribution requires seed metrics; without them, routes may not be redistributed correctly.
- Documentation consistency matters because interface and VLAN description mismatches can slow troubleshooting.

9 Conclusion

The project successfully demonstrates the design and implementation of a multi-service network in Cisco Packet Tracer. The final topology separates departments using VLANs, connects the main office and branch through WAN links, supports centralized DHCP with two DNS services, provides web service access, enables wireless clients on VLAN 40, and allows inside networks to use PAT through the main office edge router.

The verification results confirm that inter-VLAN routing, DHCP relay, dynamic routing, route redistribution, NAT/PAT, and wireless access operate correctly. The project also highlights important operational lessons, especially the need to verify ACL syntax, configure helper addresses, define redistribution metrics, and maintain accurate documentation across router and switch interfaces.